

Elementary Problems Functional Equation

Version 1.0.1

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Basic Work

1. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+2) = 2x^2 - 3x + 1$ for any $x \in \mathbb{R}$.
2. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(3x-1) = 4x + 8$ for any $x \in \mathbb{R}$.
3. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x^2+1) = x^2 + 2$ for any $x \in \mathbb{R}$.
4. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $xf(x) = (x+2)(f(x)+1)$ for any $x \in \mathbb{R}$.
5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $2xf(x) = 3f(x) - 1$ for any $x \in \mathbb{R}$.
6. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $4x^2f(x) = f(x)$ for any $x \in \mathbb{R}$.
7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x)^2 = 3f(x) + 4$ for any $x \in \mathbb{R}$.
8. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) + 3f(1-x) = -4x + 10$ for any $x \in \mathbb{R}$.
9. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f(x) + 2xf\left(\frac{1}{x}\right) = x + 2$$

for any $x \in \mathbb{R}^+$.

10. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $2f(x-1) + 3f(1-x) = 20$ for any $x \in \mathbb{R}$.
11. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + f\left(\frac{x-1}{x}\right) = \frac{x^2 + x - 1}{x}$$

for any nonzero real number x .

12. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(xy) = xf(x) + yf(y)$ for any $x, y \in \mathbb{R}$.
13. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x+y)) = f(x) + y$ for any $x, y \in \mathbb{R}$.
14. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x-y) = f(x) + f(y) - 2xy$ for any $x, y \in \mathbb{R}$.
15. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x^2 - y^2) = x^2 - f(y^2)$ for any $x, y \in \mathbb{R}$.
16. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x)f(y) - f(x+y) = x + y$ for any $x, y \in \mathbb{R}$.
17. (IMO shortlist 1969) Find all functions f defined for all x that satisfy the condition $xf(y) + yf(x) = (x+y)f(x)f(y)$, for all x and y . Prove that exactly two of them are continuous.
18. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x+y) + xy = f(x)f(y)$ for any $x, y \in \mathbb{R}$.

19. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(y) - yf(x) = f\left(\frac{y}{x}\right)$$

for any $x, y \in \mathbb{R}$ with $x \neq 0$.

20. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(xf(x) + f(y)) = x^2 + y$ for any $x, y \in \mathbb{R}$.

21. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(y + f(x))) = f(x + y) + f(x) + y$ for any $x, y \in \mathbb{R}$.

22. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(y) + f(x + f(y)) = y + f(f(x) + f(f(y)))$ for any $x, y \in \mathbb{R}$.

23. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + f(y + z)) + f(f(x + y) + z) = 2y$ for any $x, y, z \in \mathbb{R}$.

24. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + f(y)) = 2f(xf(y))$ for any $x, y \in \mathbb{R}$.

25. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(2f(f(x)) + y) = 2f(x + 3) + f(3y - 1)$ for any $x, y \in \mathbb{R}$.

26. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x + y) - x)f(f(x + y) - y) = xy$ for any $x, y \in \mathbb{R}$.

Cauchy Equation

27. (Cauchy equation) Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(x + y) = f(x) + f(y)$ for any $x, y \in \mathbb{Q}$.

28. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + y) = f(x) + f(y)$ for any $x, y \in \mathbb{R}$ and $f(x) \geq 0$ for any $x \geq 0$.

29. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ for any $x, y \in \mathbb{R}$.

30. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(x + y) + 3 = f(x) + f(y)$ for any $x, y \in \mathbb{Q}$.

31. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(x + y) + 2xy = f(x) + f(y)$ for any $x, y \in \mathbb{Q}$.

32. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x) + f(y) = 2f\left(\frac{x + y}{2}\right)$$

for any $x, y \in \mathbb{Q}$.

33. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(2f(x) + f(y)) = 2x + y$ for any $x, y \in \mathbb{Q}$.

Integers and Rationals

34. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(m + nf(m)) = m + f(m)f(n)$ for any $m, n \in \mathbb{Z}$.
35. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(m + n)^2 = f(m^2) + f(n^2)$ for any $m, n \in \mathbb{Z}$.
36. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(m - n + f(n)) = f(m) + f(n)$ for any $m, n \in \mathbb{Z}$.
37. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(n) + f(3m - n) = 2f(m)$ for any $m, n \in \mathbb{Z}$.
38. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(m + n) = f(m)f(n)f(mn)$ for any $m, n \in \mathbb{Z}$.
39. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(m + n) + f(mn - 1) = f(m)f(n) + 2$ for any $m, n \in \mathbb{Z}$.
40. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(m + n) + f(mn) = f(m)f(n) + 1$ for any $m, n \in \mathbb{Z}$.
41. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(m) + f(n)) = m + n$ for any $m, n \in \mathbb{N}$.
42. (IMO shortlist 1980) The function f is defined on the set $\mathbb{Q}$ of all rational numbers and has values in $\mathbb{Q}$. It satisfies the conditions $f(1) = 2$ and $f(xy) = f(x)f(y) - f(x + y) + 1$ for all $x, y \in \mathbb{Q}$. Determine f .
43. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(x + f(y)) = f(x) + f(y) + 2y$ for any $x, y \in \mathbb{Q}$.
44. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(f(x) + y) = x + f(y)$ for any $x, y \in \mathbb{Q}$.
45. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that $f(x + f(y)) = y + f(x + 1)$ for any $x, y \in \mathbb{Q}$.

Injective and Surjective Function

46. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x + y + f(y)) = f(f(x)) + 2y$ for any $x, y \in \mathbb{R}$.
47. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x + y) - x + y) = 2y$ for any $x, y \in \mathbb{R}$.
48. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x)) + xf(x) = 1$ for any $x \in \mathbb{R}$.
49. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x) + y) = 2x + f(f(y) - x)$ for any $x, y \in \mathbb{R}$.
50. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(xf(x) + f(y)) = f(x)^2 + y$ for any $x, y \in \mathbb{R}$.
51. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x) + xf(y)) = xf(y + 1)$ for any $x, y \in \mathbb{R}$.

52. (Kyrgyzstan MO 2012 Problem 4) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)^2 + f(y)) = xf(x) + y, \quad \forall x, y \in \mathbb{R}.$$

53. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(xf(z) + y) = zf(x) + y$ for any $x, y, z \in \mathbb{R}$.

54. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(y + zf(x)) = f(y) + xf(z)$ for any $x, y, z \in \mathbb{R}$.

Record

v1.0: problems 1–54